

PROJECT: BEE-1012

sub-system: EXP

Version: 1.0.0



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 01_COVER PAGE.SchDoc	Sheet 1 of 33
Checked by: BAO BUI Q.	Document No: 1	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025

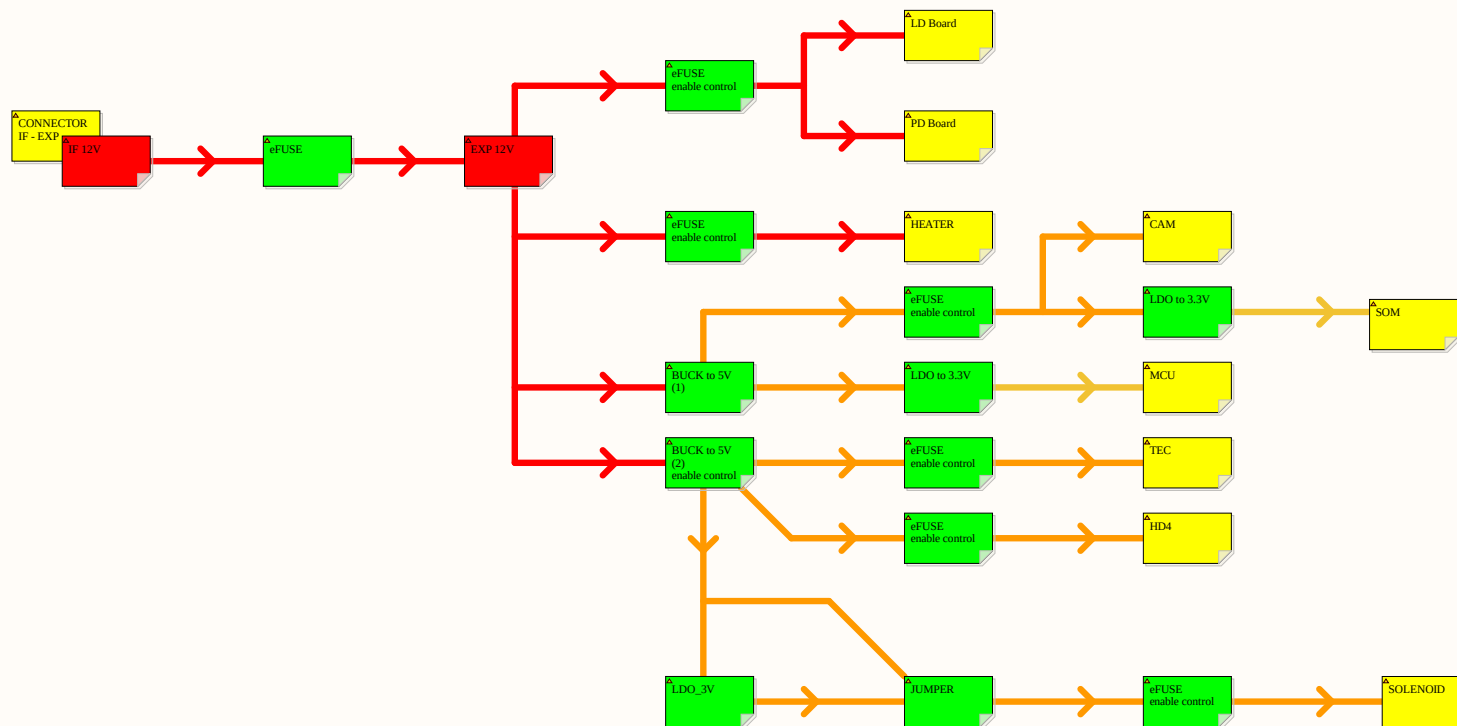
	1	2	3	4	5	6																								
A																														
B																														
C																														
D	 <table><tr><td>Project: BEE-1012</td><td colspan="5">Sub-system: SATELLITE_EXP_CONTROLLER</td></tr><tr><td>Design by: SANG HUYNH</td><td colspan="2">Sheet title: 02_BLOCK DIAGRAM.SchDoc</td><td colspan="3">Sheet 2 of 33</td></tr><tr><td>Checked by: BAO BUI Q.</td><td colspan="2">Document No: 2</td><td colspan="3">Size: A3</td></tr><tr><td>Approved by: VU PHAM</td><td colspan="2">Revision: V1.2.2</td><td colspan="3">Date: 12/6/2025</td></tr></table>						Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER					Design by: SANG HUYNH	Sheet title: 02_BLOCK DIAGRAM.SchDoc		Sheet 2 of 33			Checked by: BAO BUI Q.	Document No: 2		Size: A3			Approved by: VU PHAM	Revision: V1.2.2		Date: 12/6/2025		
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER																													
Design by: SANG HUYNH	Sheet title: 02_BLOCK DIAGRAM.SchDoc		Sheet 2 of 33																											
Checked by: BAO BUI Q.	Document No: 2		Size: A3																											
Approved by: VU PHAM	Revision: V1.2.2		Date: 12/6/2025																											
	1	2	3	4	5	6																								

INDEX

Page	Title	Page	Title	Page	Title
1	COVER PAGE	11		21	
2	BLOCK DIAGRAM	12		22	
3	INDEX	13		23	
4	PI HAT	14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

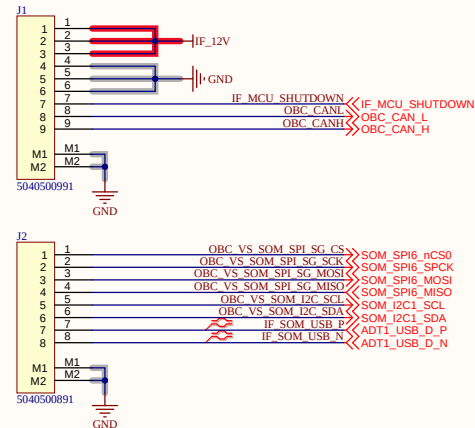


Project: BEE-1012		Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 33	
Checked by: BAO BUI Q.	Document No: 3	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025	

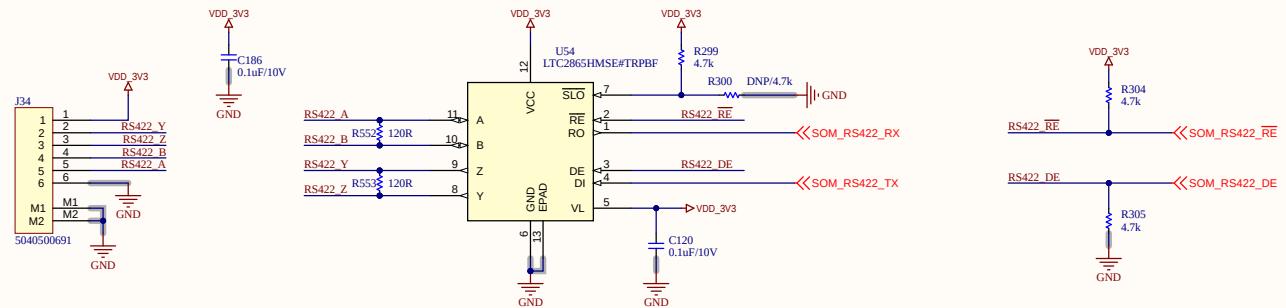


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 04_POWER_TREE.SchDoc	Sheet 4 of 33
Checked by: BAO BUI Q.	Document No: 4	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/6/2026

form IF (OBC Board)

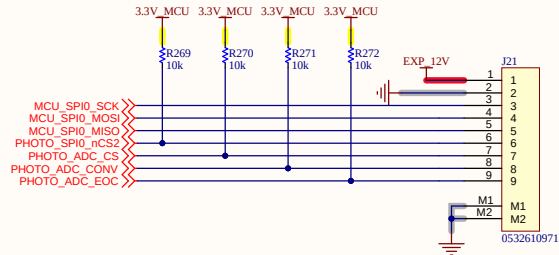


RS422

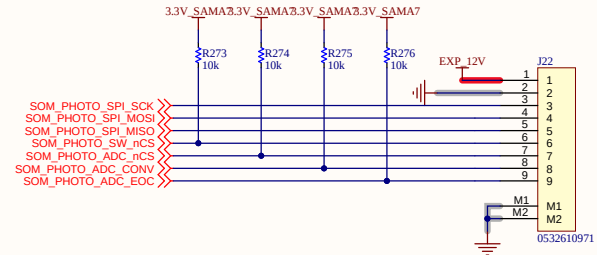


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_IN.SchDoc	Sheet 5 of 33
Checked by: BAO BUI Q.	Document No: 5	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/12/2026

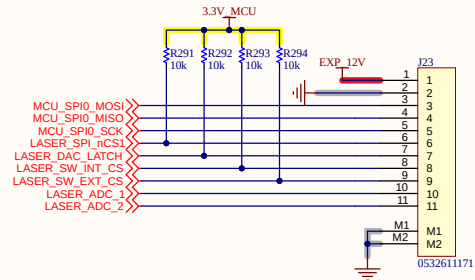
for PHOTO board



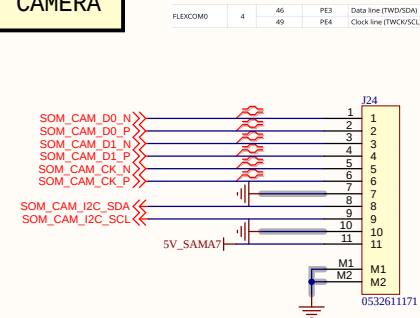
SOM to PHOTO



for LASER board



for CAMERA



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 05_CONNECTOR_OUT.SchDoc Sheet 6 of 33

Checked by: BAO BUI Q.

Document No: 6

Size: A3

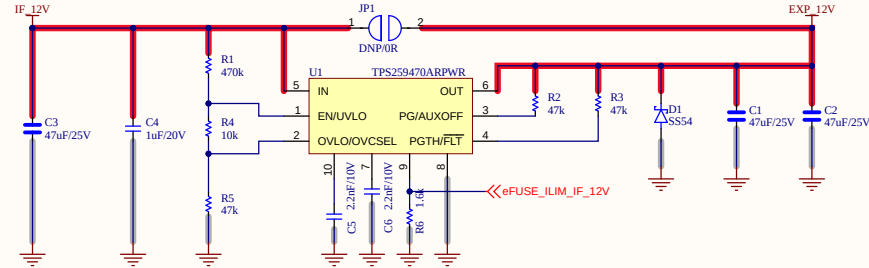
Approved by: VU PHAM

Revision: V1.2.2

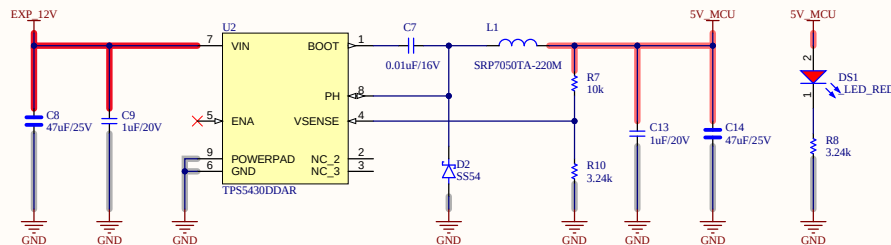
Date: 1/7/2026

EFUSE 12V

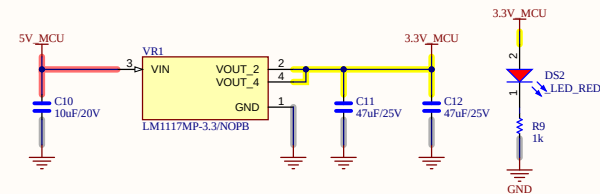
Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (47 + 470) / 47 = 13.2V$
 Slew Rate: $2000 / 2200 = 0.91V/ms$
 I-LIM: $3334 / 1600 = 2.08A$



BUCK for MCU, SOM

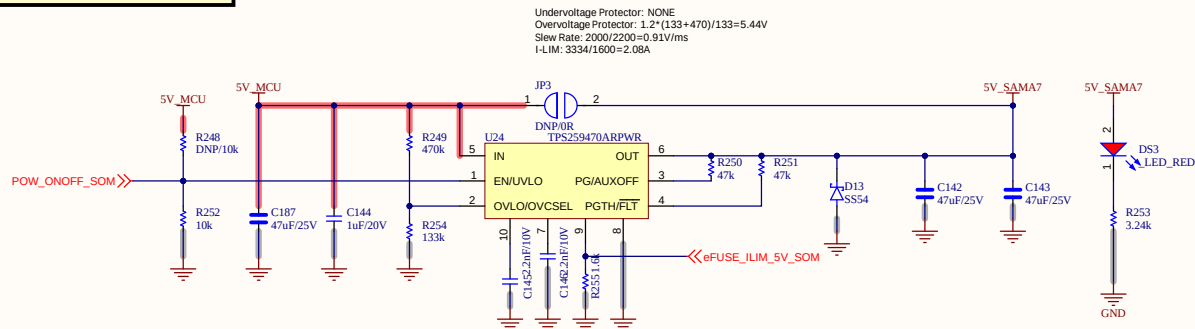


LDO for MCU

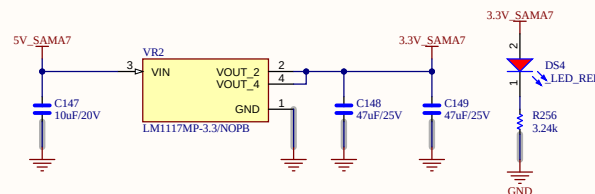


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_1.SchDoc	Sheet 7 of 33
Checked by: BAO BUI Q.	Document No: 7	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

EFUSE for SOM



LD0 for SOM



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_2.SchDoc	Sheet 8 of 33
Checked by: BAO BUI Q.	Document No: 8	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

BUCK for PERIPHERAL

The schematic diagram illustrates a BUCK converter circuit for a peripheral. The input is a 12V supply (EXP_12V) connected to a 47uF/25V capacitor (C159) to ground. The input is then connected to the VIN pin of a TPS5430DDAR buck converter (U35). The converter's BOOT pin is connected to VIN, and its PH pin is connected to a 0.01uF/16V capacitor (C158) to ground. The ENA pin is connected to a 5V_MCU supply through a 10k resistor (R411) and to a red LED (DS5, LED_RED) through a 3.2k resistor (R413). The VSENSE pin is connected to a 5V_PERI supply through a 10k resistor (R412) and to ground through a 3.2k resistor (R415). The output of the converter is connected to a 5V_PERI supply through an inductor (L6, SRP7050TA-220M) and a 1uF/20V capacitor (C161) to ground. The output is also connected to a 5V_PERI supply through a 47uF/25V capacitor (C162) to ground. The output is connected to a 5V_PERI supply through a 3.2k resistor (R413) and to ground through a 3.2k resistor (R415). The output is connected to a 5V_PERI supply through a 3.2k resistor (R413) and to ground through a 3.2k resistor (R415). The output is connected to a 5V_PERI supply through a 3.2k resistor (R413) and to ground through a 3.2k resistor (R415).

[illegible]

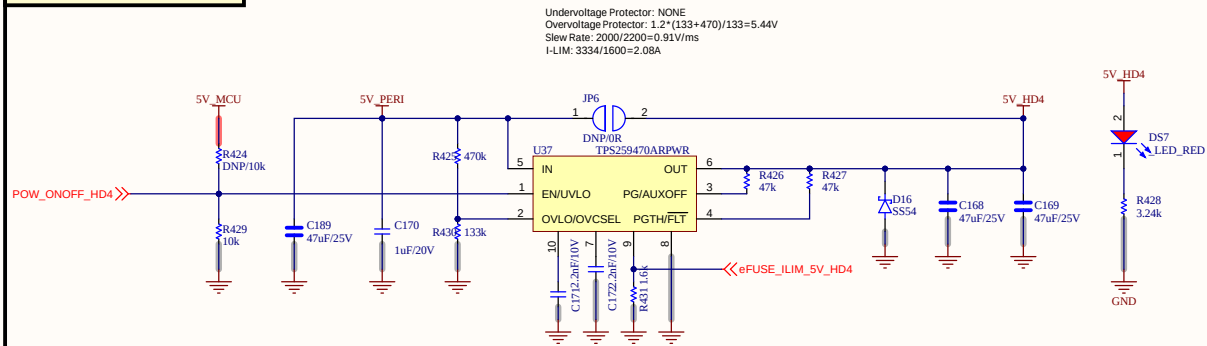
Sub-system: SATELLITE_EXP_CONTROLLER

Sheet 9 of 33

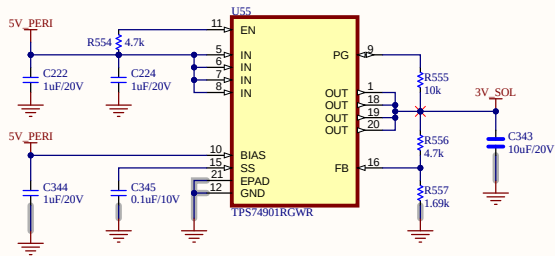
Size: A3

Date: 1/7/2026

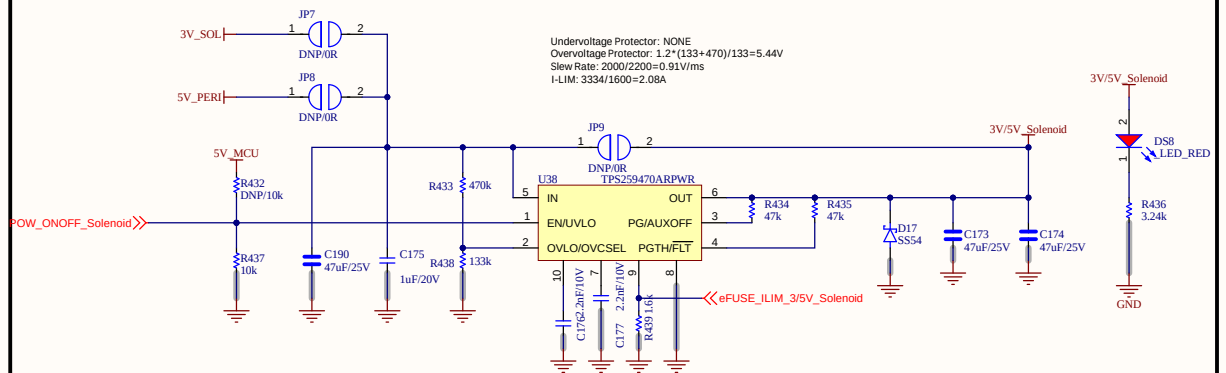
EFUSE for HD4



LDO 3V



EFUSE for SLN



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 06_POWER_4.SchDoc

Sheet 10 of 33

Checked by: BAO BUI Q.

Document No: 10

Size: A3

Approved by: VU PHAM

Revision: V1.2.2

Date: 1/8/2026

EFUSE for LP

Undervoltage Protector: NONE
Overvoltage Protector: $1.2 \times (133 + 470) / 133 = 5.44V$
Slew Rate: $2000 / 2200 = 0.91V/ms$
I-LIM: $3334 / 1600 = 2.08A$

The diagram illustrates the EFUSE for LP circuit. Key components and connections include:

- 5V_MCU** supply connected to **R128** (DNP/10k) and **R139** (10k).
- EXP_12V** supply connected to **C194** (47uF/25V), **C195** (1uF/20V), and **R204** (133k).
- LP_12V** supply connected to **C192** (47uF/25V) and **C193** (47uF/25V).
- U43** (TPS259470ARPWR) IC with pins: **IN**, **EN/UVLO**, **PG/AUXOFF**, **OUT**, **OVLO/OVCSSEL**, **PGTH/FLT**.
- JP10** (DNP) connected to **U43** pins 1 and 2.
- R192** (2mF/10V), **C197** (2mF/10V), and **R209** (1k) connected to **U43** pins 10, 7, and 9 respectively.
- R136** (47k) and **R137** (47k) connected to **U43** pins 3 and 4 respectively.
- D18** (SS54) diode connected to **U43** pin 6.
- D59** (LED_RED) connected to **LP_12V** and **GND**.
- R138** (1k) connected to **LP_12V** and **GND**.

The red line indicates the **EFUSE_ILIM_12V_LP** signal path, which is connected to **U43** pin 9.

EFUSE for HEATER

Undervoltage Protector: NONE
 Overvoltage Protector: $1.2 \times (133 + 470) / 133 = 5.44\text{V}$
 Slew Rate: $2000 / 2200 = 0.91\text{V/ms}$
 I-LIM: $3334 / 1600 = 2.08\text{A}$

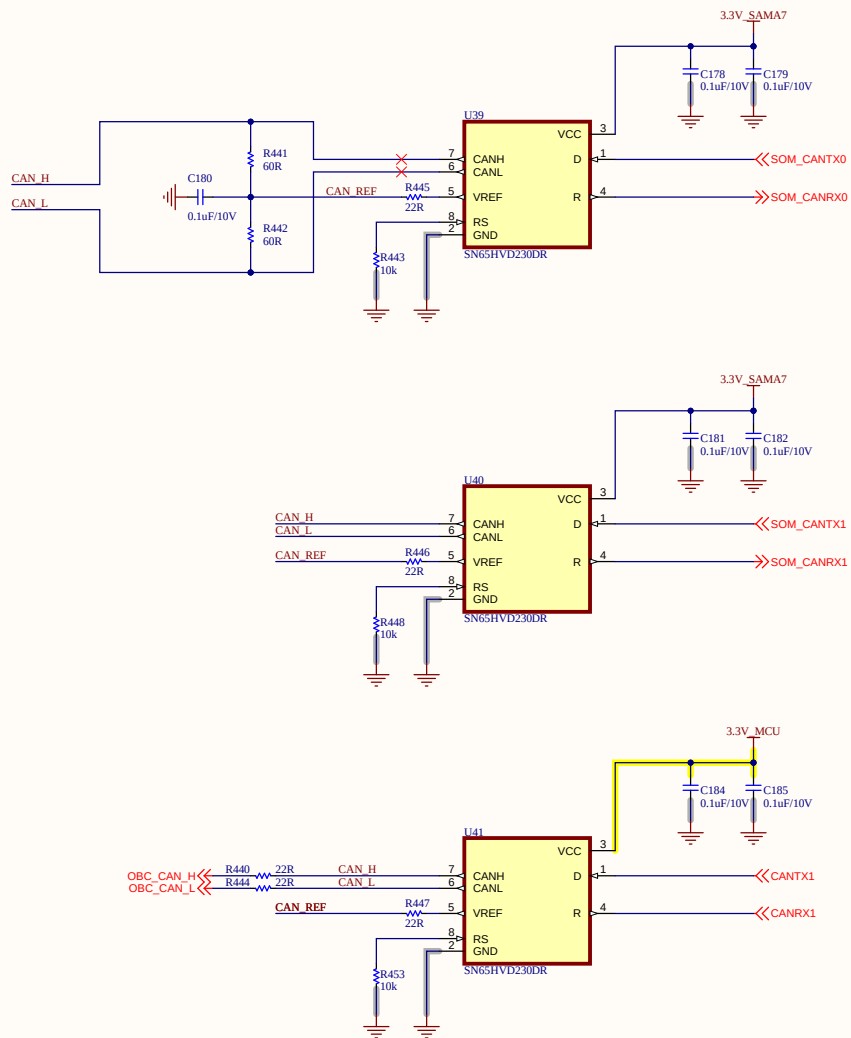


Sub-system: SATELLITE_EXP_CONTROLLER

Sheet 11 of 33

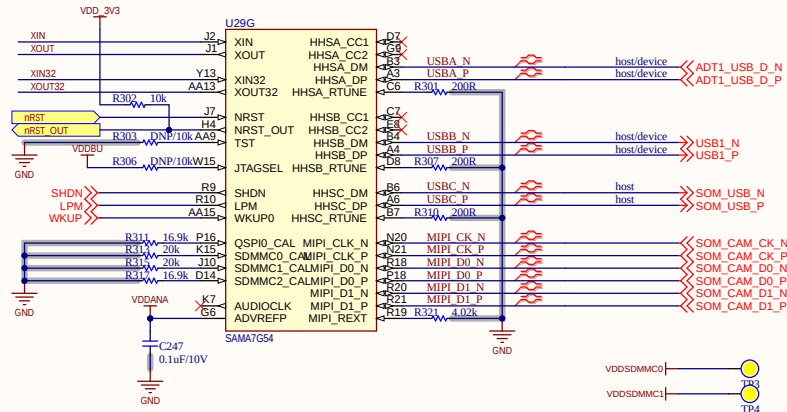
Size: A3

Date: 1/7/2026

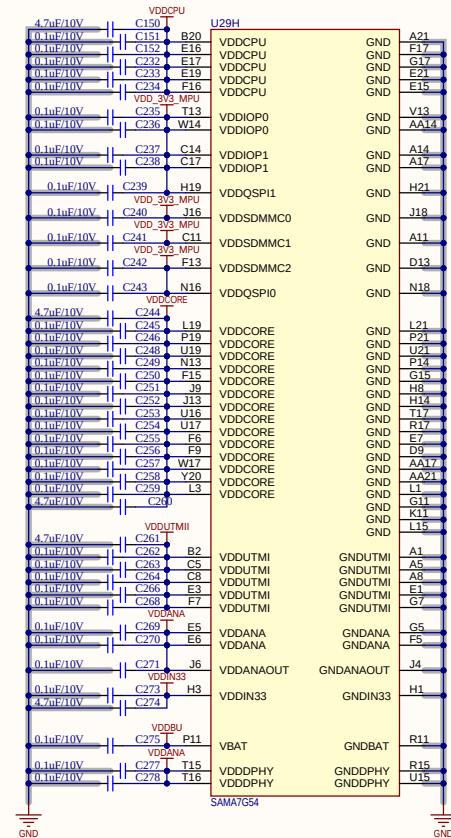


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 07_CAN.SchDoc	Sheet 12 of 33
Checked by: BAO BUI Q.	Document No: 12	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/9/2026

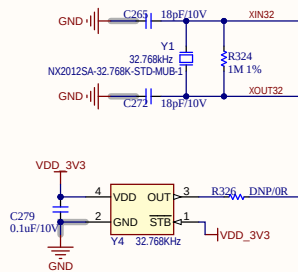
MPU configuration



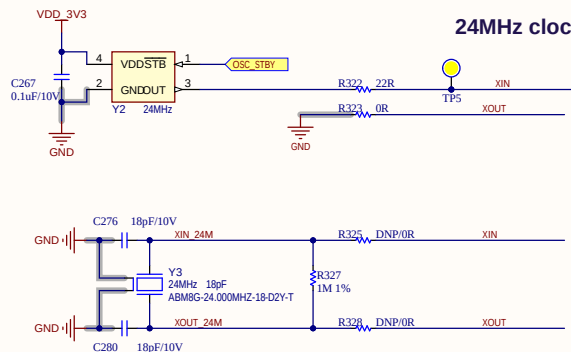
MPU decoupling



32.768KHz clock

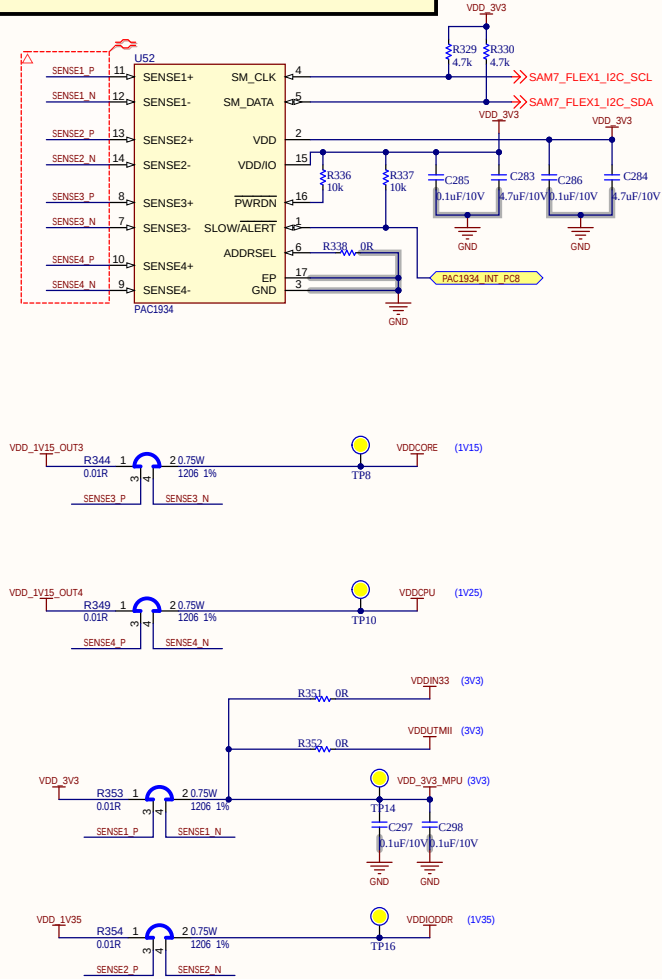


24MHz clock

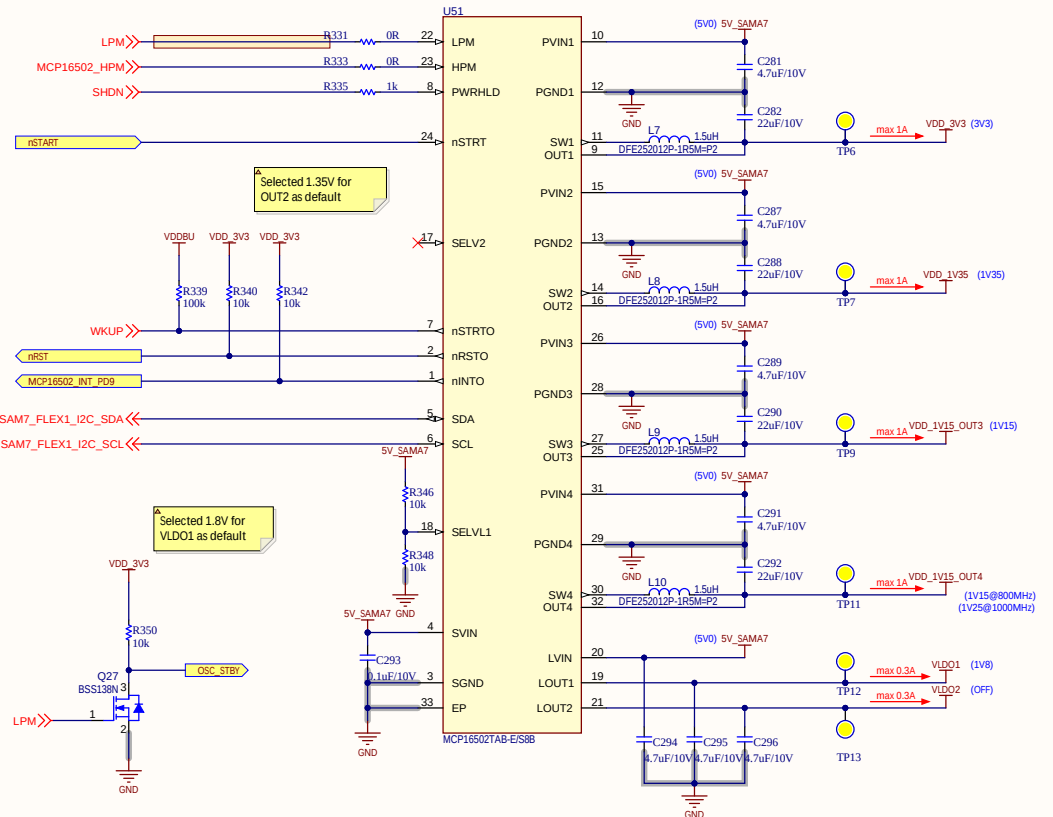


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 08_SAMA7G54-MPU.Sch	Sheet 13 of 33
Checked by: Bao Bui	Document No: 13	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

Power Measurement and Distribution

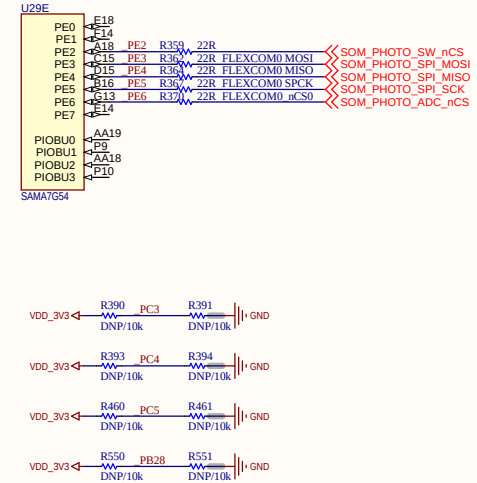


PMIC



Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 08_SAMA7G54-PMIC.Sch	Sheet 14 of 33
Checked by: Bao Bui	Document No: 14	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

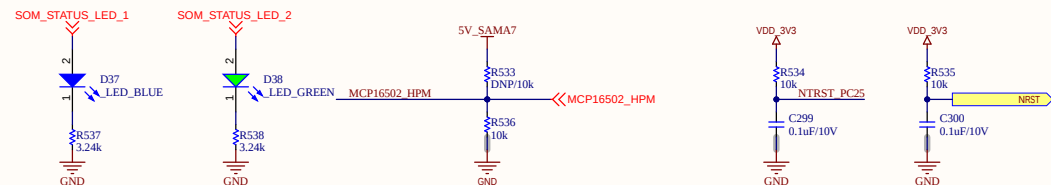
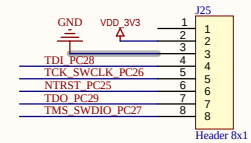
SAMA7G5 Bus Distribution



SPI:3 F0: PE3456
 F5: PA26-789
 F6

UART:3 F4:PD18 PD19
 F3:PD16 PD17
 F2:PC11 12

I2C:2 F1:PC9 10
 F8:PC1314
 F7:PC7 8



Project: BEE-PC	Sub-system: EXP
Design by: Cam Doan	Sheet title: 08_SAMA7G54-PIO.SchDoc
Checked by: Bao Bui	Document No: 15
Approved by: Vu Pham	Revision: 1.0.0
	Size: A3
	Date: 10/29/2025

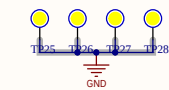
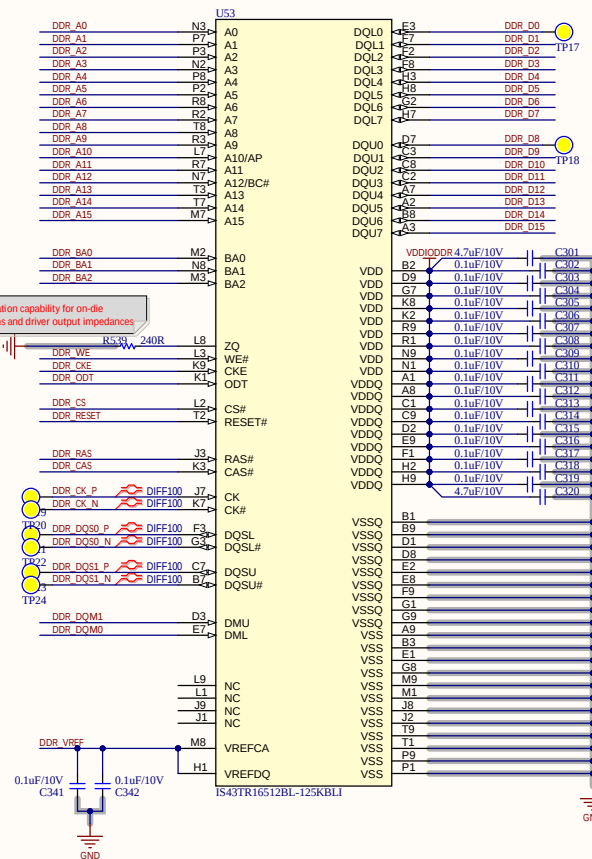
ADDDR-CTL Matched Net Lengths [Tolerance = 10mils]

SAMA7/G54 U29F

LANE0 Matched Net Lengths [Tolerance = 10mils]

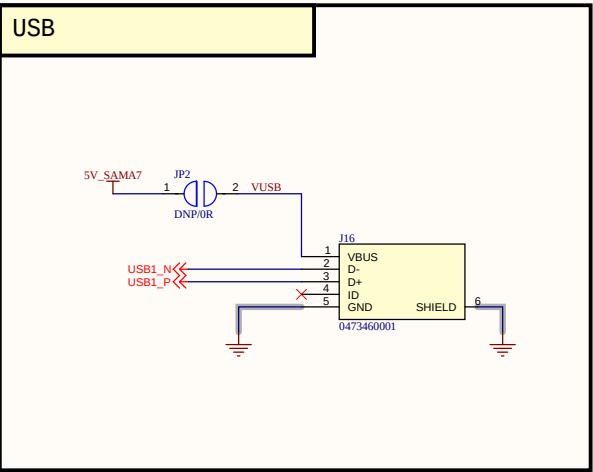
LANE1 Matched Net Lengths [Tolerance = 10mils]

RZQ: calibration capability for on-die terminations and driver output impedances

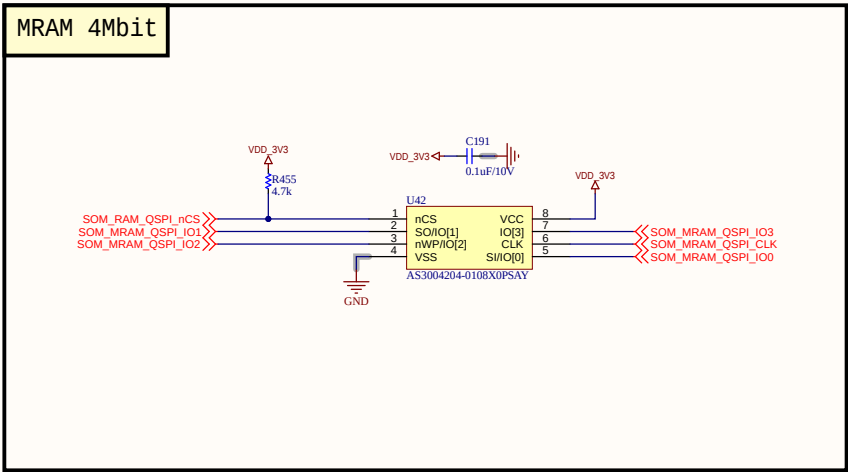


Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 08_SAMA7G54-DDR3.Sch	Sheet 16 of 33
Checked by: Bao Bui	Document No: 16	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

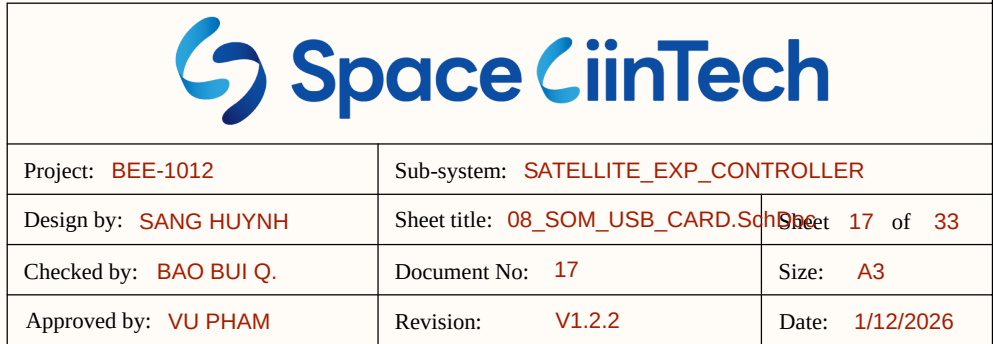
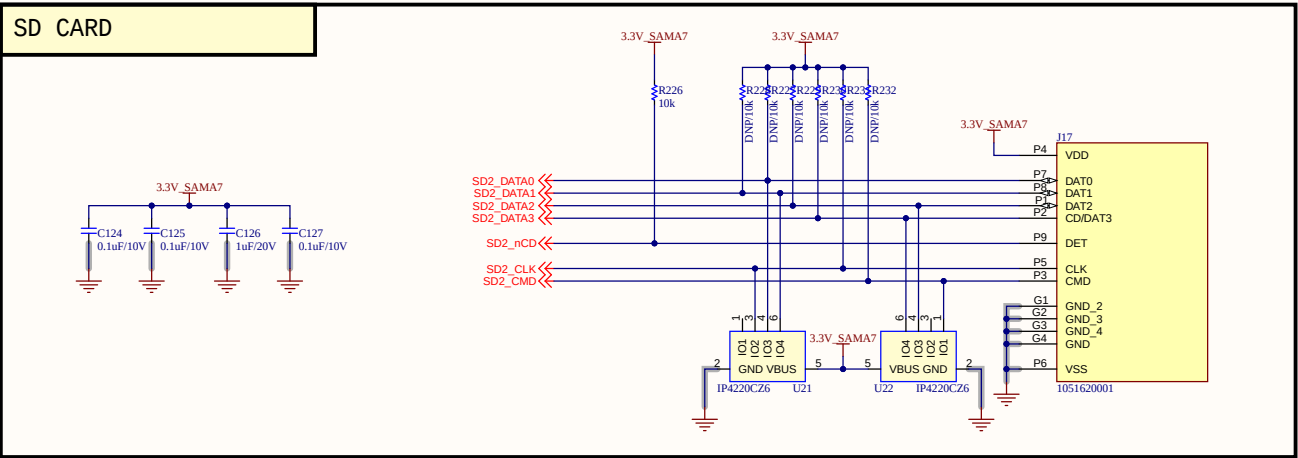
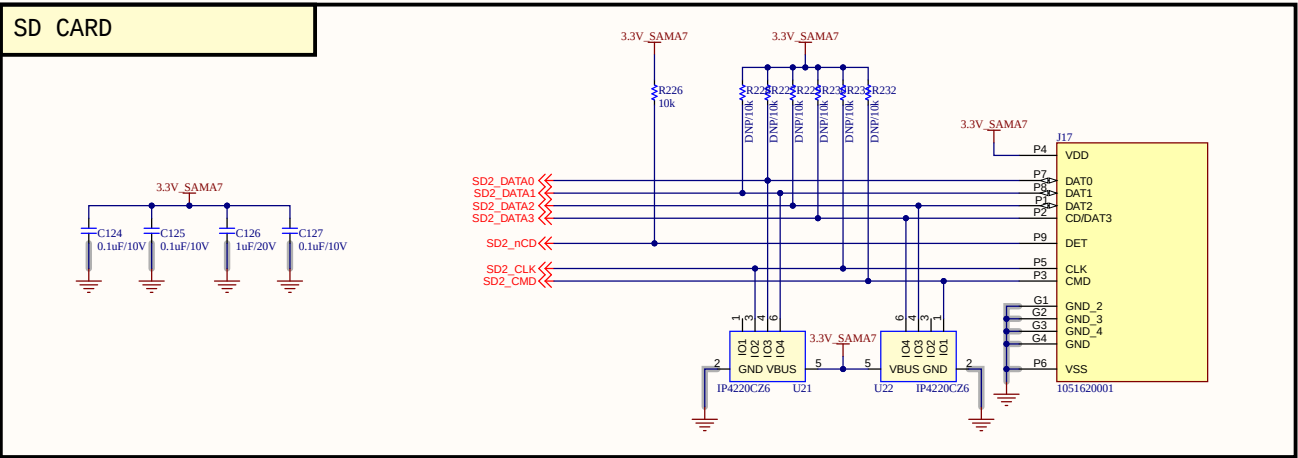
USB



MRAM 4Mbit



SD CARD



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 08_SOM_USB_CARD.SchDoc	Sheet 17 of 33
-------------------------------------	----------------

Checked by: **BAO BUI Q.**

Document No: 17

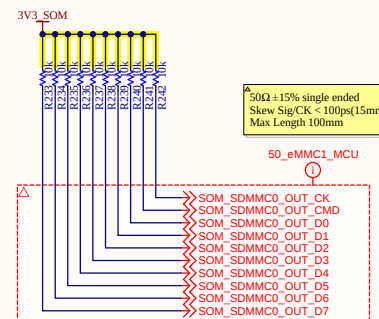
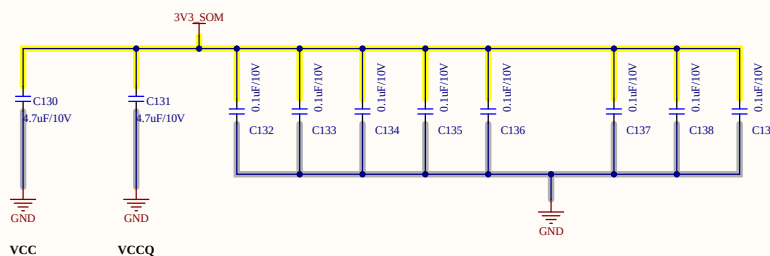
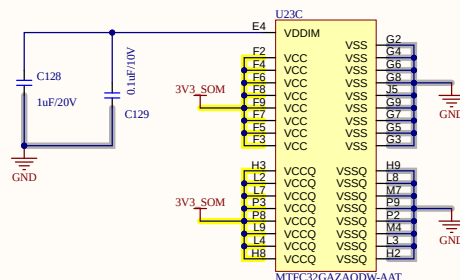
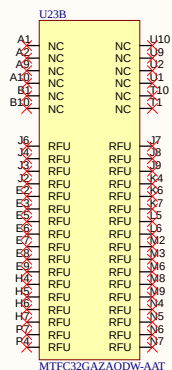
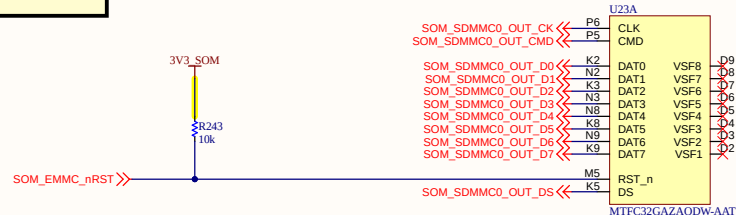
Size: A3

Approved by: **VU PHAM**

Revision: V1.2.2

Date: 1/12/2026

eMMC - Buffer



50Ω ±15% single ended
Skew Sig/CK < 100ps(15mm)
Max Length 100mm

50_eMMC1_MCU



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 09_eMMC.SchDoc

Sheet 18 of 33

Checked by: BAO BUI Q.

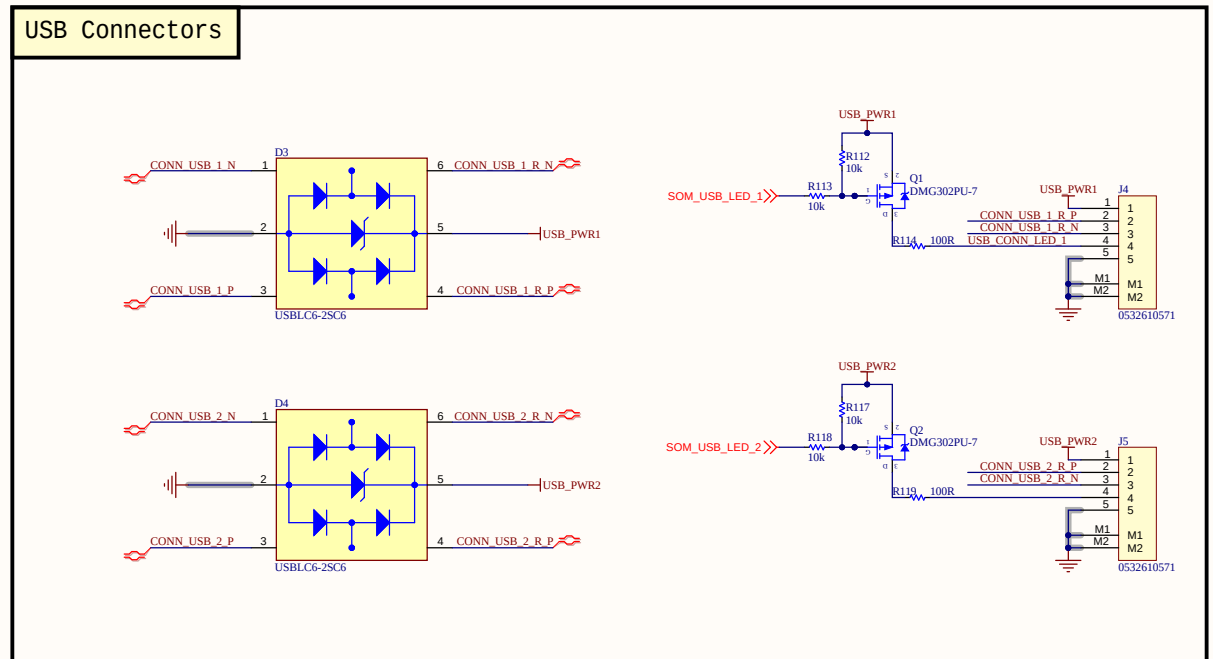
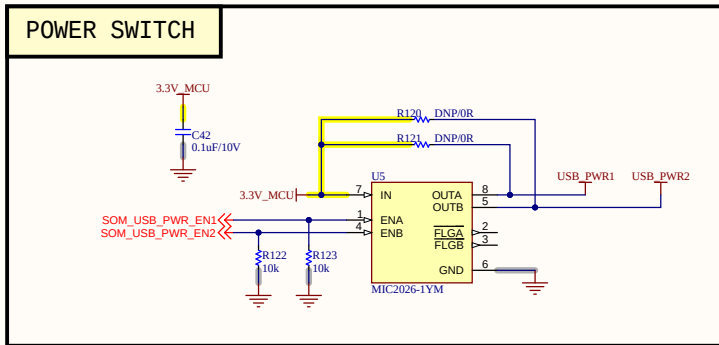
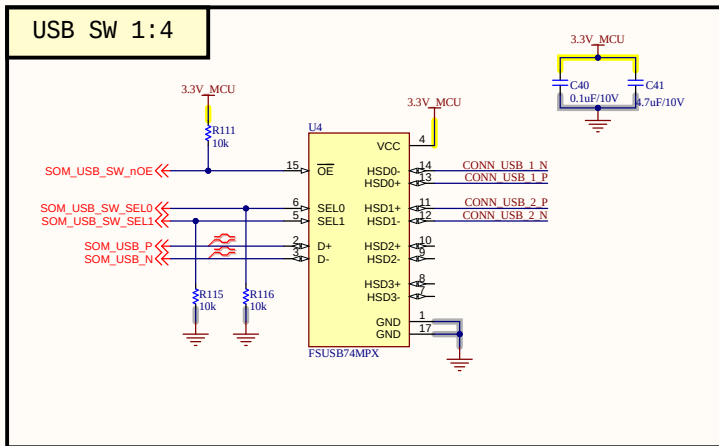
Document No: 18

Size: A3

Approved by: VU PHAM

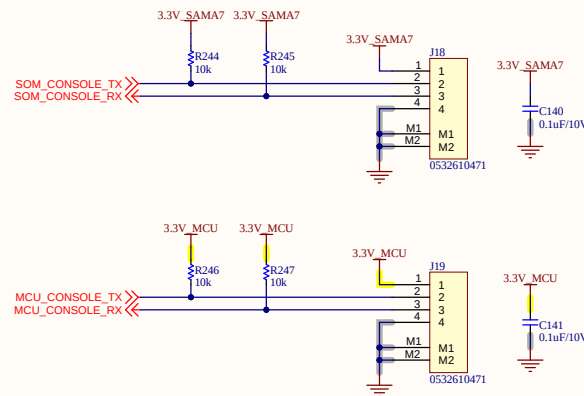
Revision: V1.2.2

Date: 1/7/2026



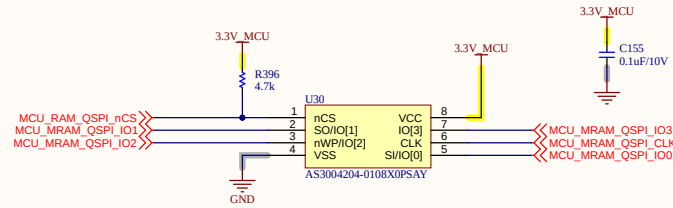
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 10_USB.SchDoc	Sheet 19 of 33
Checked by: BAO BUI Q.	Document No: 19	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

CONSOLE

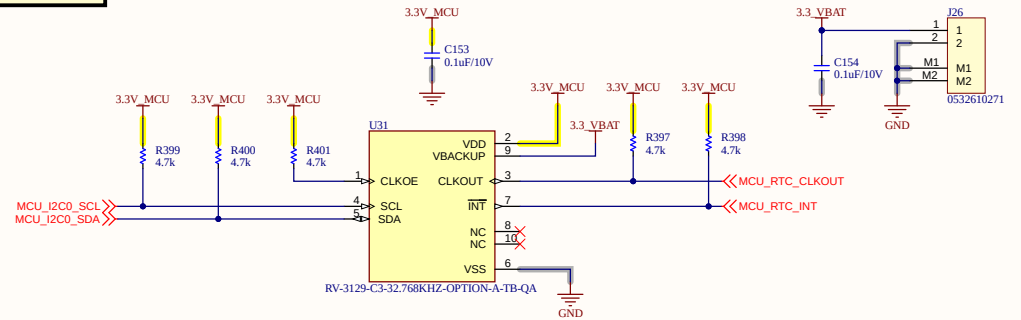


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 12_CONSOLE.SchDoc	Sheet 21 of 33
Checked by: BAO BUI Q.	Document No: 21	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

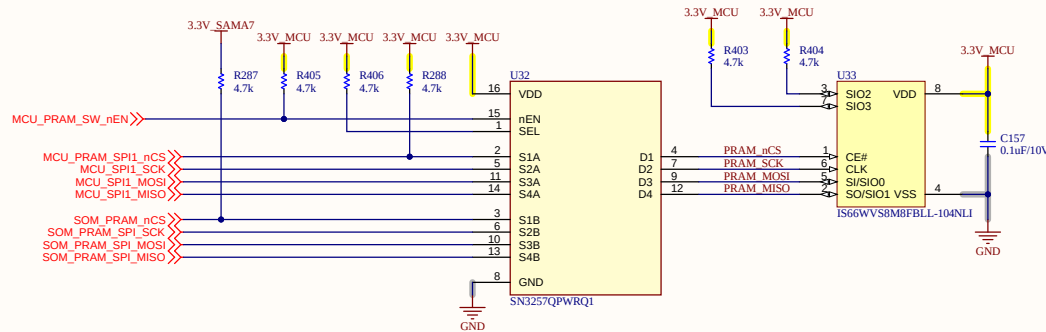
MRAM 4Mbit



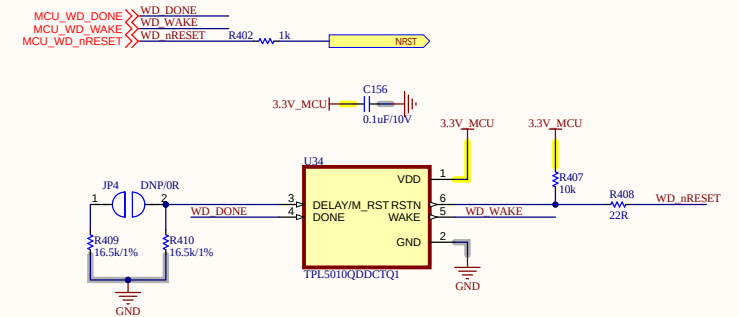
RTC



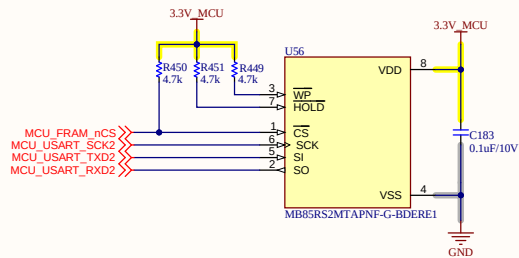
PRAM



WATCH DOG

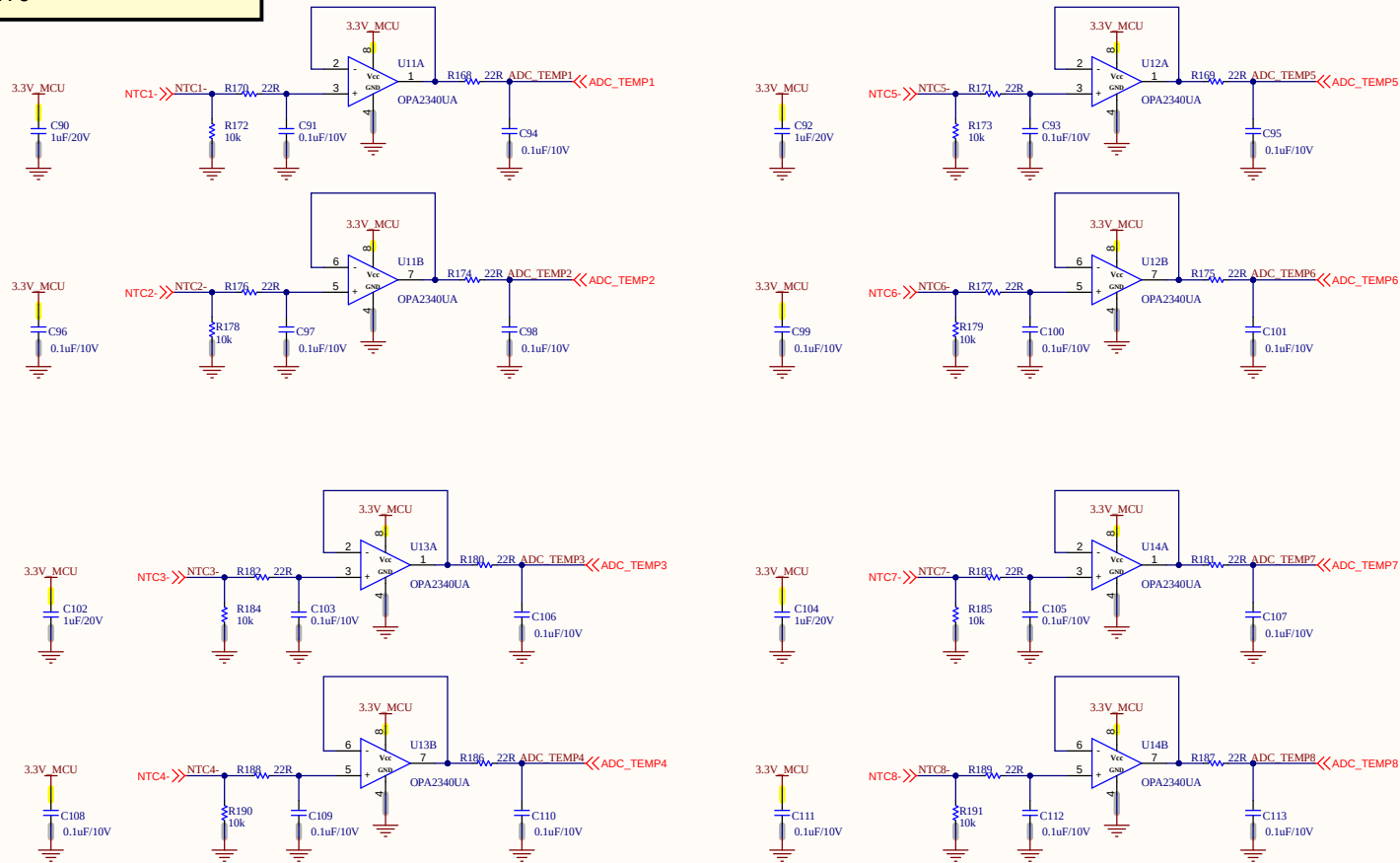


FRAM

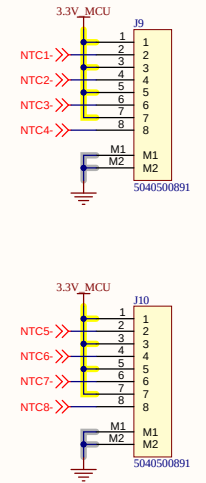


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 13_MRAM_RTC_WDT.Sch Sheet	22 of 33
Checked by: BAO BUI Q.	Document No: 22	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/9/2026

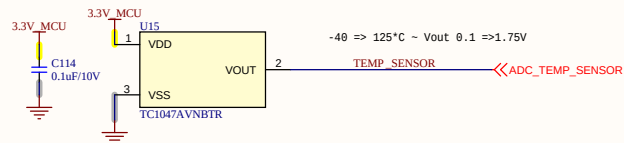
NTC



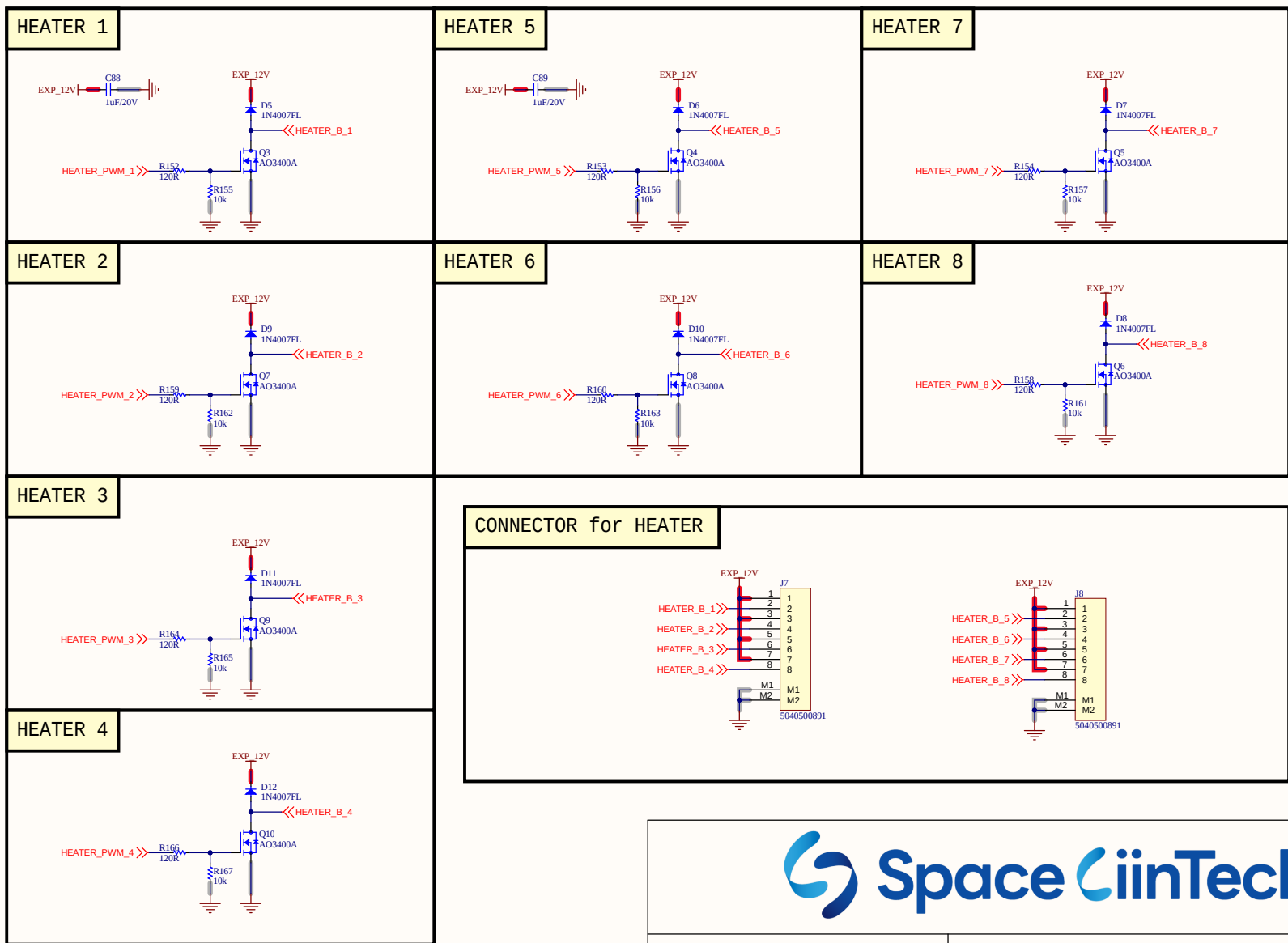
CONNECTOR FOR NTC



TEMPERATURE SENSOR

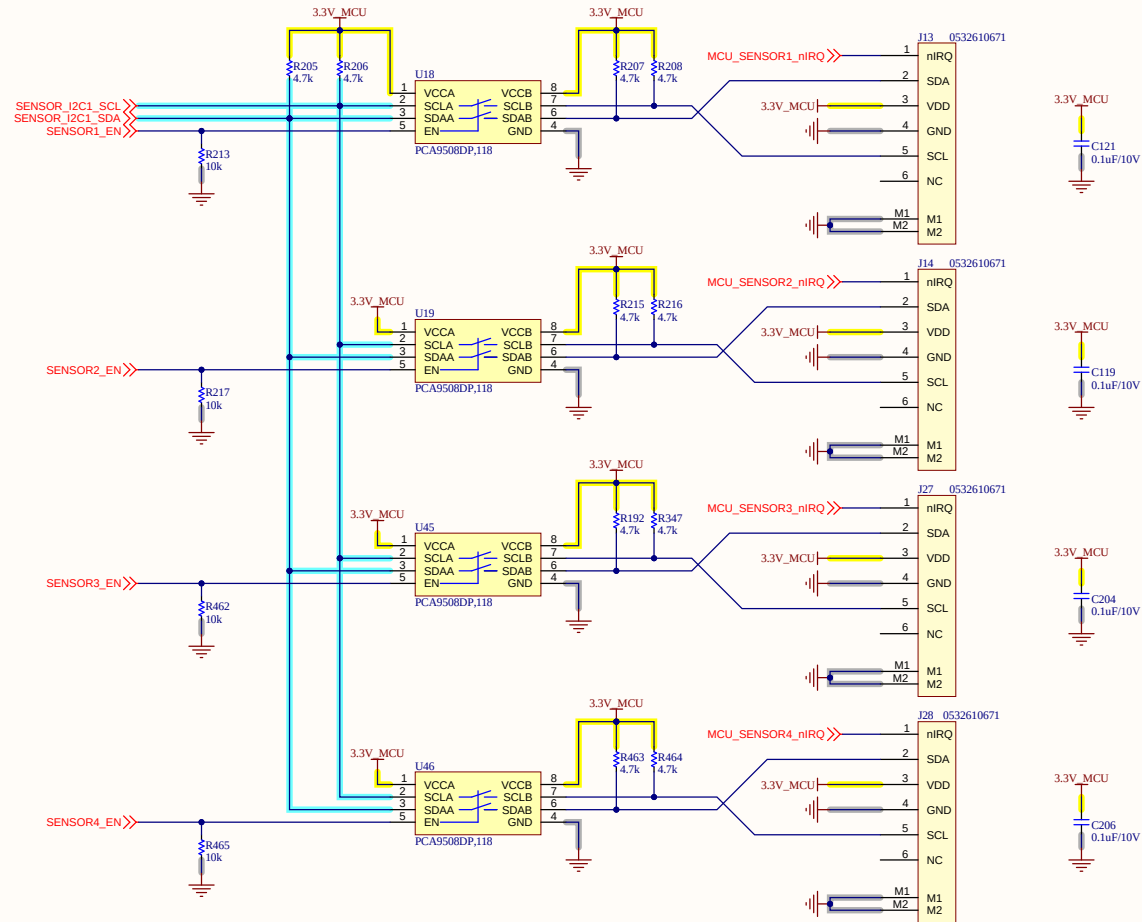


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 14_NTC_TEMPSENSOR_SCHDoc 23 of 33	
Checked by: BAO BUI Q.	Document No: 23	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

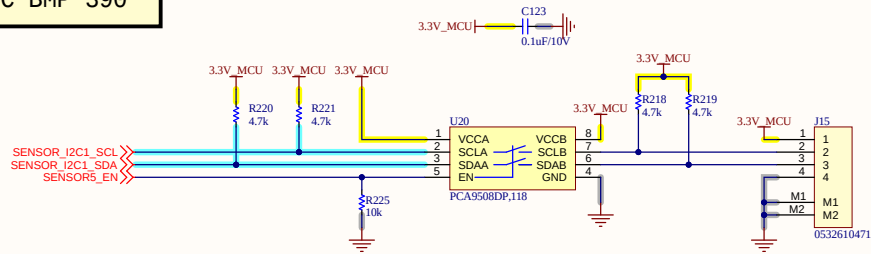


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 15_HEATER.SchDoc	Sheet 24 of 33
Checked by: BAO BUI Q.	Document No: 24	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

I2C

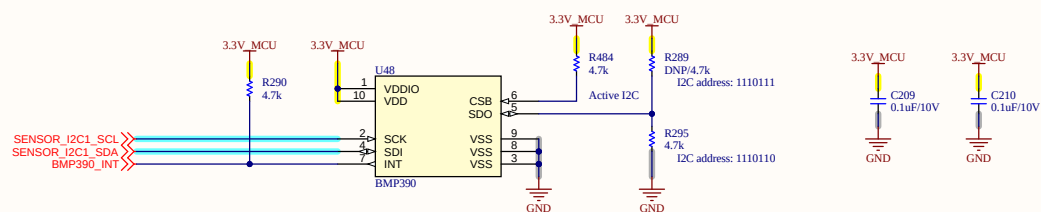


I2C BMP 390

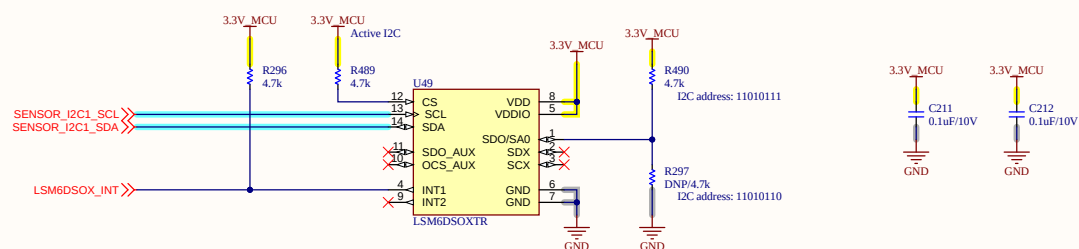


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 17_FLOW_Sensor.SchDocSheet 26 of 33
Checked by: BAO BUI Q.	Document No: 26
Approved by: VU PHAM	Revision: V1.2.2
	Size: A3
	Date: 1/12/2026

BMP390



LSM6DS0X



Project: BEE-1012

Sub-system: SATELLITE_EXP_CONTROLLER

Design by: SANG HUYNH

Sheet title: 17_I2C_Sensor_OnBoard.SchDoc27 of 33

Checked by: **BAO BUI Q.**

Document No: 27

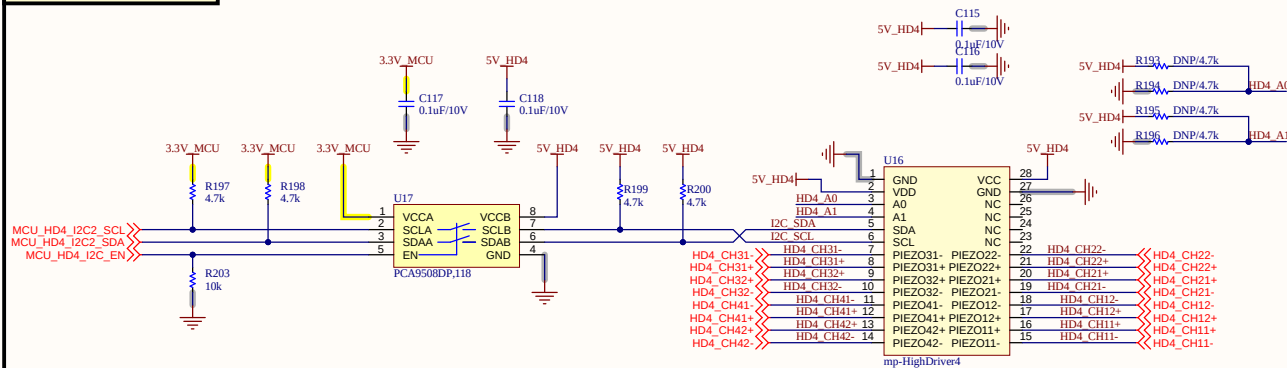
Size: A3

Approved by: **VU PHAM**

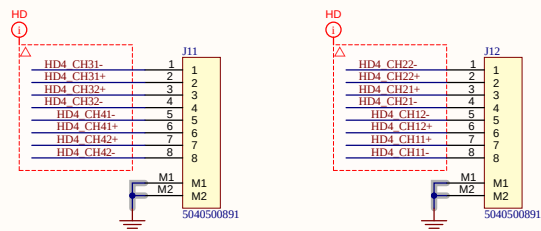
Revision: V1.2.2

Date: 1/12/2026

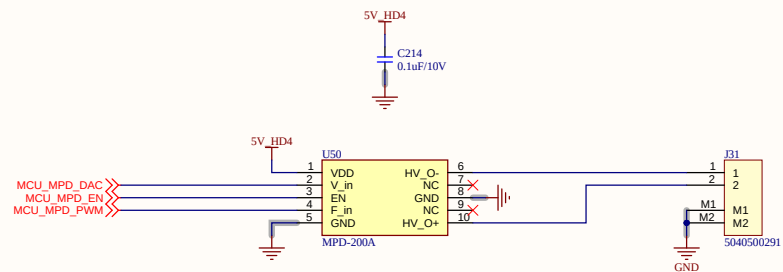
HIGH DRIVER 4



CONNECTOR BUMP

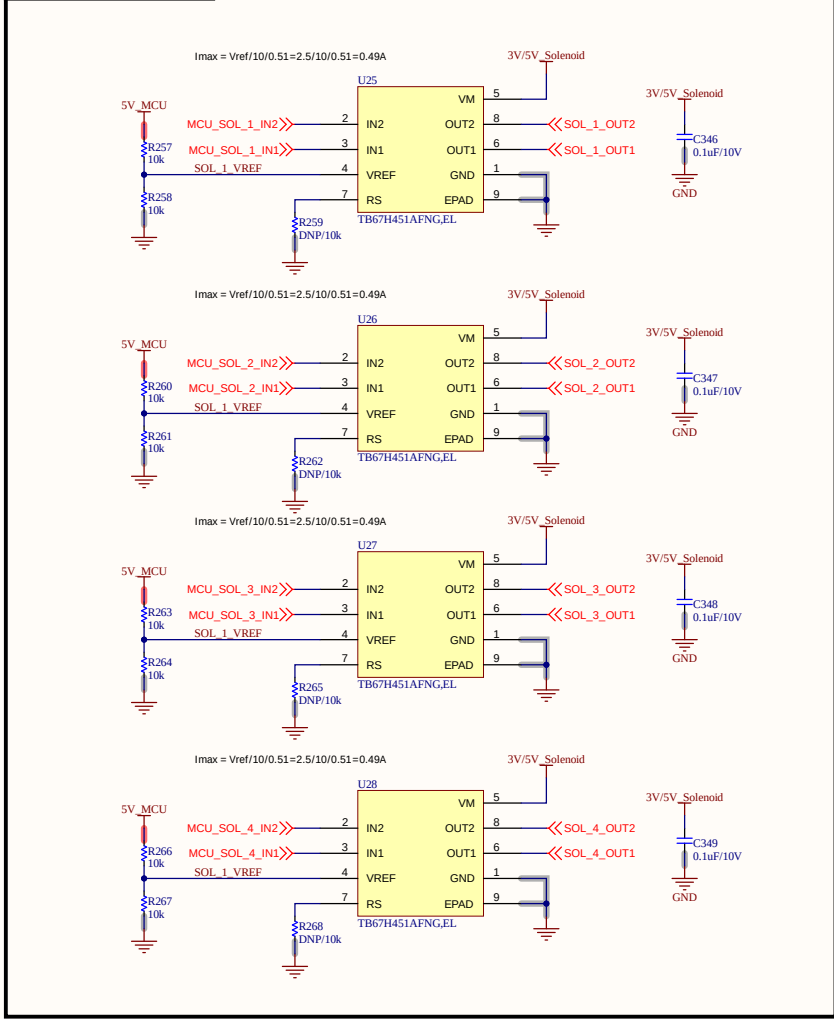


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 18_HD4_PUMP.SchDoc	Sheet 28 of 33
Checked by: BAO BUI Q.	Document No: 28	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

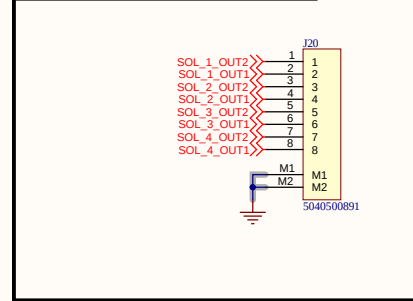


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 18_HD4_PUMP_2.SchDoc	
Checked by: BAO BUI Q.	Document No: 29	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/6/2026

SOLENOID

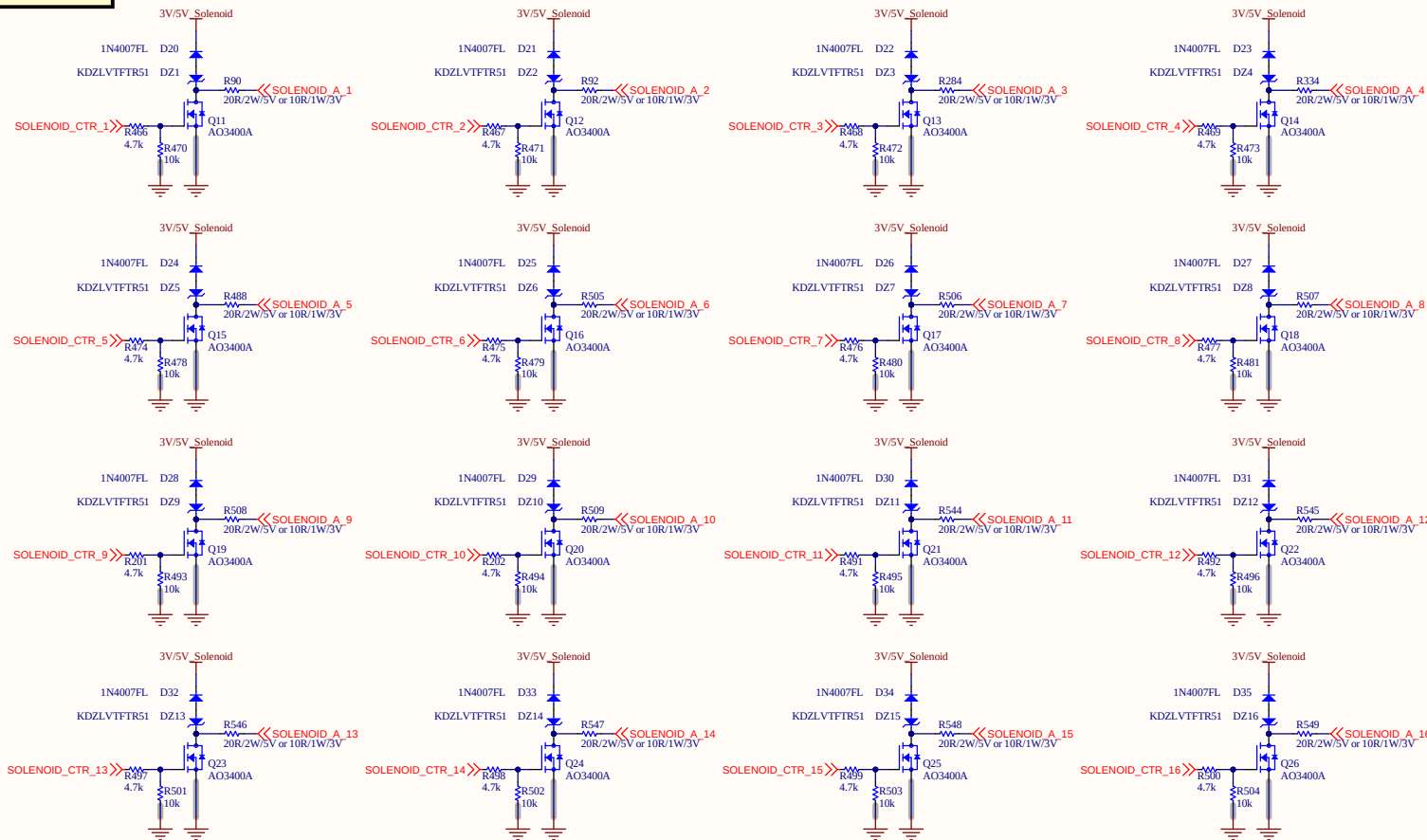


CONNECTOR SOLENOID

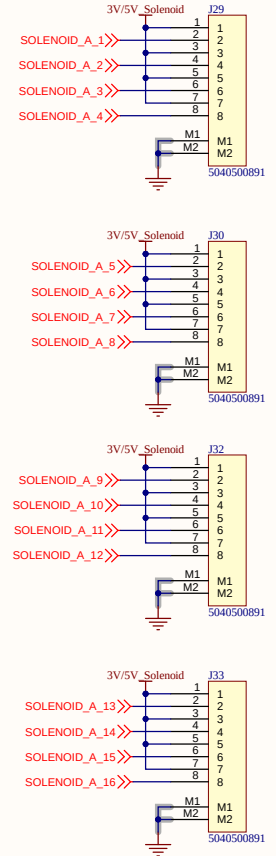


Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 19_SOLEDNOID_1.SchDoc	Sheet 30 of 33
Checked by: BAO BUI Q.	Document No: 30	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/8/2026

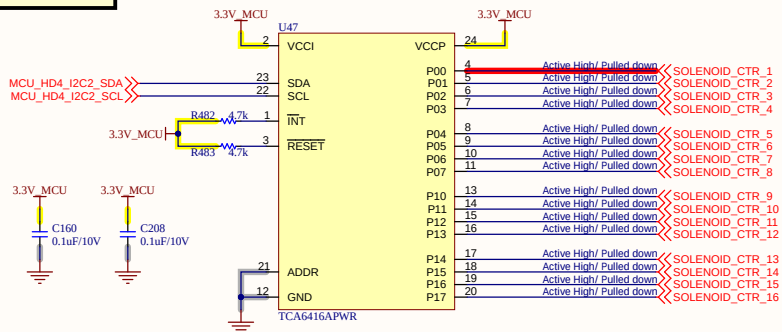
SOLENOID



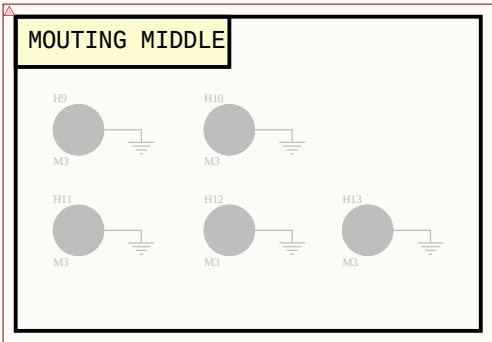
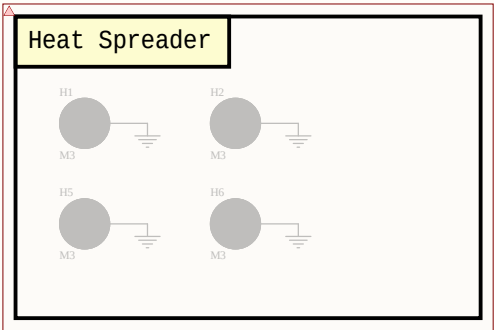
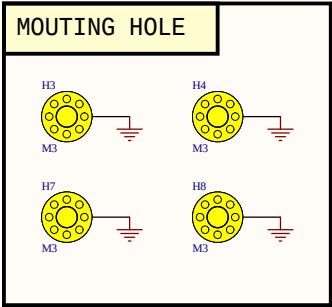
CONN SOLENOID



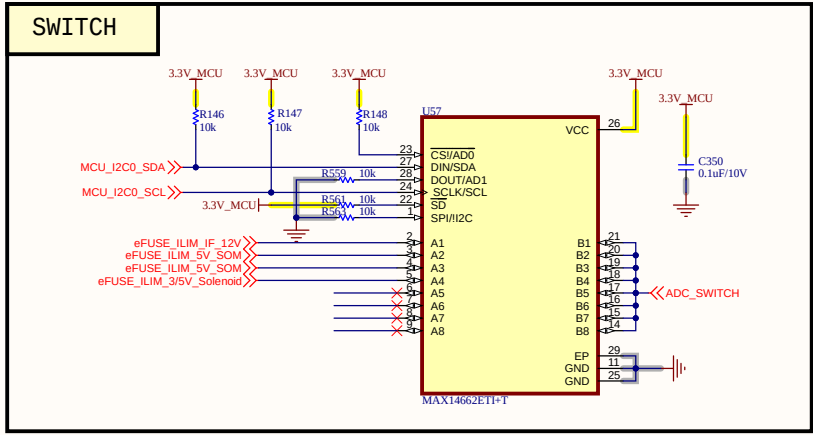
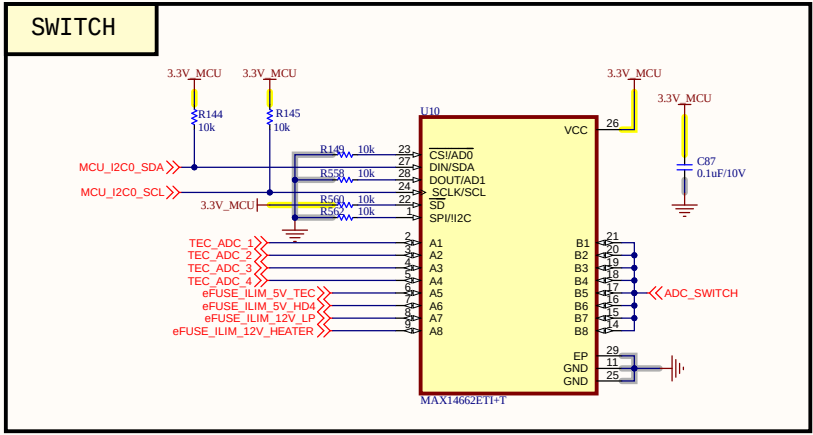
I/O EXPANDED



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 19_SOLEDNOID_2.SchDoc	Sheet 31 of 33
Checked by: BAO BUI Q.	Document No: 31	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: MOUTING.SchDoc	Sheet 32 of 33
Checked by: BAO BUI Q.	Document No: 32	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/26/2025



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: ANA_SWITCH.SchDoc	Sheet 33 of 33
Checked by: BAO BUI Q.	Document No: 33	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/12/2026